

BIOL 2011: Environmental Issues

CRN: 20048

3 CREDIT HOURS

COURSE DATES: Fall 2025

MWF 10 – 10:50 AM, Mathile 311

Instructor: Leslie Riley

Office Hours: MW 11 AM – 1 PM; F 11AM – 12 PM & as needed

Office Location: Meyer 118

Telephone: (419) 772 – 3143 (my office)

E-mail: l-riley.1@onu.edu

CATALOG DESCRIPTION

The interaction of humans and their environment with an emphasis on the problems arising from increasing human population, pollution, and resource use.

COURSE PREREQUISITE(S)

N/A

GENERAL EDUCATION

4: Knowledge of the physical/natural world

COURSE READINGS AND OTHER MATERIALS

- Karr, S. 2021. Environmental Science for a Changing World (4th edition). W. H. Freeman and Company: New York, NY
 - Achieve with e-book only: 9781319496739
- OR
- Achieve with loose-leaf book: 9781319570224
- (If you want to purchase from MacMillan directly:
<https://store.macmillanlearning.com/us/product/Scientific-American-Environmental-Science-for-a-Changing-World-Digital-Update/p/1319496644>. If you go this route, make sure you purchase the e-book AND Achieve Access.
- Additional homework assignments and course materials will be available on both Achieve and Canvas.

ACHIEVE

Within the course Canvas page, you will find the link for the textbook website and Achieve materials. For any Achieve homework you will need to go to this website. For the first log in, please follow these directions:

- Go to <https://achieve.macmillanlearning.com/courses/bemrh4/> to access Achieve for this course. If you already have a Macmillan Learning account, you should be able to log in with your existing credentials and skip to the next step. If you do not yet have an account, you will be prompted to create an account.
- After you are signed in, click on the box on the top right, “Enroll in a new course.”
- Enter your access code that you received from the bookstore or MacMillan.
- You may also need to enter the course ID: bemrh4
- Click “Finish enrollment.”
- To access your e-book, expand the sidebar on the left side of the screen. Scroll down to “e-book.”
- **Need Help?** Check out additional directions here for joining a course:
<https://mhe.my.site.com/macmillanlearning/s/article/Achieve-Join-a-course>
- You can also contact their tech support team:
<https://macmillan.force.com/macmillanlearning/s/contactsupport>

- If you are still having trouble, please let me know and we can troubleshoot together.

COURSE OUTCOMES

Upon successful completion of this course, students will be able to:

- Articulate why environmental literacy, scientific literacy and information literacy are necessary skills to address environmental problems and pursue sustainable solutions
- Describe basic principles in ecology, evolution and conservation biology and demonstrate how these principles can be used to mitigate ecosystem damage, inform restoration efforts, preserve biodiversity and predict species responses to changing environmental conditions
- Evaluate how human population size and per capita impacts can affect sustainability and environmental health
- Describe how economics and policy can affect management of selected natural resources
- Demonstrate how anthropogenic activities can affect the quantity and quality of freshwater resources, and the health of marine ecosystems
- Compare sustainable farming and fishing techniques to industrial farming and fishing techniques and evaluate the costs and benefits associated with each
- Evaluate fossil fuels and compare the environmental and health costs of each
- Describe types and sources of air pollution and explain how air pollution impacts the environment and human health

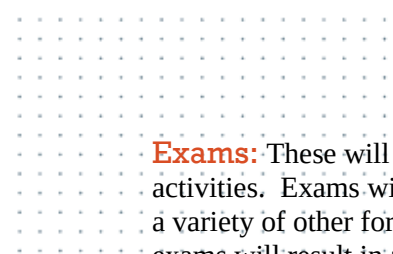
COURSE POLICIES

Attendance: To ensure that you get the most out of this course, plan to attend and actively participate in all virtual and face-to-face meetings. This includes participating in discussions in person and in online forums. These activities will all have a small number of points associated with them, but can add up if you miss these activities. If you should miss a virtual or face-to-face meeting, it is your responsibility to contact me about missed announcements or upcoming assignments. If you have an approved university excuse to miss class (e.g. athletics, band, drama, field trip, etc.), it is your responsibility to let me know **in advance** so appropriate arrangements can be made. There will be **no** make-up quizzes, exams or in-class activities without proper prior notification.

Evaluation: Your final grade will be based upon the following criteria:

• Exam #1 (Monday, Sept. 29)	40
• Exam #2 (Monday, Nov. 3)	60
• Final Exam (Friday, Dec. 19 @ 9:15 AM)	75
• Reflection Paper (Monday, Nov. 10)	15
• Team Projects (Dec. 5, 8, 10)	20
• Pre-Assessments, Achieve Assignments, Canvas Assignments, Online Forums & In-Class Activities	140
Total Points	350

Grading Scale: A: 90 – 100%, B: 80 – 89%, C: 70 -79%, D: 60 – 69%, F: 59% and below



Exams: These will cover material from assigned readings, lectures, supplemental videos and other class activities. Exams will be in person. Exams will consist of multiple choice questions and questions presented in a variety of other formats (figure interpretation, diagram, short answer, etc.). Unexcused absences during exams will result in a **zero**.

Reflection Paper on Volunteer Experience or Book: The purpose of this assignment is to become immersed in a local project or book related to some of the environmental issues we will discuss. You have the option to pursue a volunteer experience OR to read a book. You do not have to do both, although you are certainly more than welcome to do so. If you choose a volunteer experience, it will involve a minimum of a 3-hour time commitment, proof of your involvement and a brief reflection paper based on your experience. If you choose the book option, your reflection paper will serve as evidence that you have read the book. Further details and book options will be explained more fully on a separate handout and on Canvas.

Team Projects: Given the brief duration of the semester, there are many issues in environmental science that we can only briefly cover or may not cover at all. The team projects allow you to delve further into case studies on some of these topics. The projects will culminate in each team presenting their case study to the class in **oral presentation** format. The oral presentations will occur during the last two weeks of the semester. Details will be explained more fully on a separate handout.

Pre-Assessments: Pre-assessments will be embedded within Canvas or completed during face-to-face instruction. This provides me with important information about the level of background knowledge the whole class has with respect to particular subjects.

Achieve Assignments: After reading assigned material, taking notes from lectures and supplemental videos, and completing in-class learning activities, you will complete assignments on Achieve called **Adaptive Quizzes** to reinforce these concepts and to check your progress.

In-Class Activities: We will also complete a number of learning activities in class. You will be responsible for coming to class prepared in order to actively participate in these activities. Material to cover prior to class will be clearly stated for each in-person meeting.

Forum Participation: Many discussions for this course will happen in class, but there will also be times where discussion occurs online. When that is the case, you will participate in a discussion forum on Canvas for that particular topic.

Canvas Assignments: Some topics will also have Canvas assignments associated with them. When this is the case, assignments will have submission buttons on Canvas.

Late Work: Late work will be accepted with a deduction of 10% per day, up to one calendar week after the due date. Unless otherwise noted, **assignments are due at the BEGINNING of class or at time periods indicated online.** For late online assignments, you might need to contact me if you no longer have access to the appropriate submission button. **I will NOT accept assignments turned in by email unless I have explicitly given you permission to do so. These assignments should be submitted via Canvas or Achieve.**

Additional Expectations:

1) Come to class prepared and review material as we cover it. I expect that you are reading your textbook ahead of time, attending lectures, reviewing supplementary material and completing any pre-assessments before coming to class. This makes for more interesting and meaningful discussions during class. I also expect that you are studying material as we cover it, not just right before the exam.

2) Be proactive. If you are having trouble in the course, please come see me or contact me **EARLY** in the semester. I encourage all of you to visit office hours, set up additional appointments or email me about questions or concerns in the course. I will do my best to help in any way, but I usually cannot read minds and cannot help if I am not aware that you are having trouble.

3) Ask questions inside and outside of class. Feel free to ask me questions in class or stop by office hours. Forming study groups with your classmates is a great idea but just remember that all assignments, quizzes and exams are your own work (see link to Academic Dishonesty Policy below).

4) Be respectful to your classmates. It is imperative that everyone feels comfortable in the classroom and in online forums. Respect for diversity is expected. This includes diversity in race/ethnicity, national origin, cultural background, sexual orientation, gender identity and opinions. In addition, distractive activities will not be allowed. This includes, but is not limited to, listening to music and other inappropriate uses of electronic devices (e.g. talking on the phone, texting, etc.) during class. If you are texting, I reserve the right to finish your message for you and treat the entire class to a pop quiz on your behalf.

TENTATIVE COURSE SCHEDULE

Week	Date	Topic
Week 1	25-Aug	Introduction
	27-Aug	Environmental, Science & Information Literacy (Chapter 1)
	29-Aug	Environmental, Science & Information Literacy (Chapter 1)
Week 2	1-Sep	LABOR DAY
	3-Sep	Environmental, Science & Information Literacy (Chapter 1)
	5-Sep	Environmental, Science & Information Literacy (Chapter 1)
Week 3*	8-Sep	Ecology (Chapter 2)
		Achieve Adaptive Quizzes 1.1, 1.2 Due
	10-Sep	Ecology (Chapter 2)*
	12-Sep	Ecology (Chapter 2)*
Week 4*	15-Sep	Ecology (Chapter 2)
	17-Sep	Ecology (Chapter 2)*
	19-Sep	Ecology (Chapter 2)*
Week 5	22-Sep	Evolution & Biodiversity (Chapter 3)
		Achieve Adaptive Quizzes 2.1, 2.2, 2.3 Due
	24-Sep	Evolution & Biodiversity (Chapter 3)
	26-Sep	Evolution & Biodiversity (Chapter 3)
Week 6	29-Sep	EXAM #1 (Chapters 1 - 2)
	1-Oct	Evolution & Biodiversity (Chapter 3)
	3-Oct	Human Populations & Environmental Health (Chapter 4)

Week 7	6-Oct	Human Populations & Environmental Health (Chapter 4)
		Achieve Adaptive Quizzes 3.1, 3.2 Due
	8-Oct	Human Populations & Environmental Health (Chapter 4)
	10-Oct	Human Populations & Environmental Health (Chapter 4)
Week 8	13-Oct	FALL BREAK
	15-Oct	Managing Resources: Environmental Economics & Policy (Chapter 5)
	17-Oct	Managing Resources: Environmental Economics & Policy (Chapter 5)
		Ecological Footprint HW Due
		Achieve Adaptive Quizzes 4.1, 4.3 Due
Week 9	20-Oct	Managing Resources: Environmental Economics & Policy (Chapter 5)
	22-Oct	Managing Resources: Environmental Economics & Policy (Chapter 5)
	24-Oct	Water Resources (Chapter 6)
Week 10	27-Oct	Water Resources (Chapter 6)
		Tracking Your Trash HW Due
	29-Oct	Water Resources (Chapter 6)
	31-Oct	Water Resources (Chapter 6)
		Achieve Adaptive Quizzes 5.1, 5.2, 5.3 Due
Week 11	3-Nov	Exam #2 (Chapters 3 - 5)
	5-Nov	Food Resources (Chapter 8)
	7-Nov	Food Resources (Chapter 8)
Week 12	10-Nov	Food Resources (Chapter 8)
		Reflection Paper Due
	12-Nov	Food Resources (Chapter 8)
	14-Nov	Conventional Energy: Fossil Fuels (Chapter 9)
		Achieve Adaptive Quizzes 6.1, 6.2, 6.3 Due
Week 13	17-Nov	Conventional Energy: Fossil Fuels (Chapter 9)
	19-Nov	Conventional Energy: Fossil Fuels (Chapter 9)
	21-Nov	Air Quality & Climate Change (Chapter 10)
	Nov 24 - Nov 28	THANKSGIVING BREAK
Week 14	1-Dec	Air Quality & Climate Change (Chapter 10)
		Coal Movie HW Due
	3-Dec	Air Quality & Climate Change (Chapter 10)
	5-Dec	Team Project Presentations
		Achieve Adaptive Quizzes 8.1, 8.2, 8.3, 8.4, 9.1, 9.2 Due
Week 15	8-Dec	Team Project Presentations
		Climate Change Movie HW Due

	10-Dec	Team Project Presentations
	12-Dec	Review & Wrap-Up
		Achieve Adaptive Quizzes 10.1, 10.2 Due
Week 16	Dec 15 – Dec 19	FINAL EXAM - FRIDAY, DECEMBER 19 @ 9:15 AM – 11:15 AM

* During the Wednesdays and Fridays of weeks 3 & 4, I will be off - campus teaching a field biology course at the Metzger Nature Center. We will meet in person on Mondays of those weeks. Wednesdays and Fridays will be asynchronous and online those weeks.

COURSE FEES

N/A

TUTORING

If you need some extra help (in addition to assistance from your instructors), you can set up individual tutoring sessions through the Student Success Center. Please let me know if you need assistance with scheduling a tutor.

ONU UNIVERSAL COURSE POLICIES

Ohio Northern University is dedicated to providing an equitable educational experience for all enrolled students. Universal course policies applicable to all courses can be found at the following link: https://my.onu.edu/registrars_office/policies. Specifically, this website includes the policies for the following topics:

- Academic Dishonesty Policy
 - Academic Accommodations Policy
 - ONU Mercy Health Center Policy
 - Title IX Policy
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